Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
EIGHTH SEMESTER B.TECH (R,S) DEGREE EXAMINATION, APRIL 2025(2019 SCHEME)

Course Code: CXT418

Course Name: DESIGNING HUMAN CENTERED SYSTEMS

Max. Marks: 100		Duration: 3 Hours

PART A

		Answer all questions, each carries 3 marks.	Marks
1		A **human-centred system** is a system designed with the primary focus on human needs, abilities, and behaviours. It ensures that technology, processes, and interactions are optimized for user experience, efficiency, and accessibility. The different methods for observing human experience are: 1. Ethnographic Research 2. Participatory Research 3. Evaluative Research	(1.5) (1.5)
2		Innovators offer solutions that people don't even know they want. This cluster of methods allows you to engage with your intended audience by equipping them with creative ways to express themselves. If you pay close attention you'll discover critical and latent needs. The different methods include: • What's on Your Radar? • Buy a Feature • Build Your Own • Journaling	(1.5) (1.5)
3		A visual representation of relationships between tasks, roles, tools, and workflows. Used to identify commonalities, variations, and dependencies.	(3)
4		• Guides your plan for future research. • Documents and summarises your research findings. • Builds a shared understanding. • Deepens your empathy for others. • Challenges your preconception. • Facilitates interdisciplinary collaboration. • Communicates complex ideas visually. • Documents critical touch points.	(6 points - 3)

		• Informs subsequent design activities.	
5		• **Integrate and summarize the contextual data**. • **Point back to the data**, to maintain the "chain of custody" to ensure that the design is based on real contextual data. • **Provide a shared focus** for analysis now and, later, design. • **Provide intermediate deliverables**, which can be important to your working relationship with the customer.	(3 points 3 Mark)

6		• **Interaction Design Requirements** are a crucial step in the design process.
		• They define what a system or product needs to accomplish to meet the needs of its users and to function effectively in its intended environment.
		• This process focuses on:
		    • identifying user needs,
		    • understanding the task context, and
		    • Determining the functional and non-functional criteria for the design.
		• This involves carefully examining the Work Activity Affinity Diagram (WAAD) and any preliminary design-informing models, such as the flow model.
		• Through deductive reasoning, user needs and requirements can be extracted to construct the first span of the bridge between analysis and design.
			(3)

7		• The term "design" is used broadly to refer to the entire system development lifecycle. • It is also used narrowly to describe the creative synthesis of ideas in interaction design. • Involves understanding the audience and creating a design that meets their needs. • Involves creating and testing prototypes. • "Develop" and "development" could work but are strongly associated with software engineering and coding. • Involves the physical creation of a product or service. • Involves translating a prototype into reality.	(3 mark – 3 difference)
8		• Ideation is an active, fast-moving collaborative group process for forming ideas for design. • It is an activity that goes with design thinking; ideation is a tool of design thinking. • Iterative Exploration: Ideation is about exploration—it requires constant iteration.	(3)

		• Designers must test multiple ideas, similar to how Thomas Edison tested thousands of materials before perfecting the lightbulb. • Sketching and mockups should be made early and often, allowing users to be involved in refining the design. • Idea Creation vs. Critiquing • There are two main modes of thinking in ideation: i. Idea Creation → Generating new ideas freely, brainstorming without restrictions. ii. Critiquing → Evaluating and refining ideas based on feasibility and effectiveness.	
9		**Modern Software Prototyping Tools** Today, there are many tools designed to make prototyping **faster and more intuitive**, such as: 1. **Wireframing Tools** (for low to medium fidelity) ○ Examples: **Balsamiq, OmniGraffle, Microsoft Visio** ○ Used for **quick sketches and layout design** before adding interaction. 2. **Interactive Prototyping Tools** (for medium to high fidelity) ○ Examples: **Axure RP, Figma, Adobe XD, Sketch, InVision** ○ Allow designers to create **clickable prototypes** that simulate real interactions. 3. **Code-Based Prototyping Tools** (for high fidelity and functional prototypes) ○ Examples: **Framer, HTML/CSS/JavaScript, React, Swift, Flutter** ○ Used for **prototyping with real code**, making them closer to final implementation.	(3mark – 3 tools)
10		● Prototyping is a way to create a **design representation**, which can overlap with the actual design process. ● While it might appear that prototyping happens at a specific stage in the lifecycle, it is not limited to a particular phase. Prototyping often occurs iteratively alongside other activities. ● Interactivity in a prototype refers to how it responds to user actions, such as clicking, typing, or navigating through screens. ● The amount of interactivity is often linked to the **level of fidelity**: higher interactivity generally requires higher-fidelity prototypes.	(3)

		• Interactivity helps simulate real-world usage, making it easier to evaluate user behavior and refine designs.	
PART B			
Answer one full question from each module, each carries 14 marks.			
		Module I	
11	a)	• An approach to interviewing and observing people in their own environment. • “WHAT PEOPLE SAY, what people do, and what they say they do are entirely different things,” observed influential anthropologist Margaret Mead. • Following this wisdom, it is crucial that we pay attention to what people say and what people do in order to get a clear picture of what really happens. • A Contextual Inquiry places you in the midst of a person’s environment where you can inquire about his or her experiences in context as they are happening. Consequently, input comes directly from the people who have the most knowledge, saving you from making assumptions about how and why things are done. • Identify a location and the people to be involved. • Prepare your questions and recording equipment. • Go to the site. • Introduce yourself and the purpose. Obtain consent. • Ask the participants to do tasks in a normal way. • Observe their actions in an unobtrusive manner. • Interject questions at opportune moments. • Record your findings in videos, photos, and notes. • Thank each participant. • Benefits: • Reveals what people actually do and say • Deepens your empathy for others • Challenges your assumptions • Builds credibility with stakeholders	(7)
	b)	• An approach to conducting field research in an unobtrusive manner. • In situations where you cannot speak directly with people, or do not want to interrupt the flow of their activities, being a fly on the wall has its advantages. • You’ll come to find that careful, unobtrusive observation provides valuable insight you cannot otherwise obtain.	(7)

		• When left to their own devices, people are likely to say or do things that they're not aware of and would not be able to articulate, even if prompted. • If you can watch and listen without interfering, you have a chance to capture people's natural behavior. Remember to pay careful attention to people's tasks and workflow, taking note of the information, tools, and people they rely upon to do what they do. • Also be mindful of the surrounding environment, understanding that peripheral objects, sounds, and people may affect outcomes. • Identify a subject area to study. • Develop a plan to guide your investigation. • Consider which people and activities to watch. • Choose a location to visit. • Obtain the necessary access and permission(s). • Prepare materials for capturing what you see. • Go out and observe. • Record your findings in videos, photos, and notes. Benefits: • Diminishes your presence as a researcher • Deepens your empathy for others • Challenges your assumptions • Informs subsequent research activities	
		OR	
12	a)	• A way of building empathy for people through firsthand experience. • AS ATTICUS FINCH teaches us in To Kill a Mockingbird, "You never really understand a person until you consider things from his point of view—until you climb into his skin and walk around in it." • While any research method can help you better understand people and their needs, Walk-a-Mile forces you to take a person's journey and experience their joys, conflicts, and weariness. • In other words, you must not only see, but also feel what it is like to live in the world as someone else. • Identify whose experience you want to replicate. • Choose the tasks and activities you will perform. • Assemble what is needed to run a simulation.	(7)

		• Determine the best location. • Obtain the necessary access and permission(s). • Conduct the targeted tasks. • Do each activity as realistically as possible. • Note your findings along the way. Benefits • Helps you gain firsthand knowledge • Fosters an attitude of humility • Deepens your empathy for others • Informs subsequent research activities	
	b)	When you look at things critically, with an eye toward improvement, you set your course in the direction of making things better. These methods are good for assessing the usefulness and usability of solutions meant to serve people in new and better ways. The methods are: Think-Aloud Testing: A testing format where people narrate their experience while performing a given task. Benefits: • Reveals what people are thinking • Deepens your empathy for others • Uncovers opportunities for improvement • Lowers development costs through early discovery Heuristic Review: An auditing procedure based on ten rules of thumb for good design. Benefits • Leverages proven principles of good design • Helps you identify problems quickly • Yields data in the absence of test participants • Shows opportunities for improvement Critique: A forum for people to give and receive constructive feedback. Benefits: • Facilitates constructive discussion • Reveals blind spots in your design activities • Shows opportunities for improvement • Builds organizational alignment System Usability Scale: Scale A short survey for quantifying feedback from subjective assessments of usability. Benefits:	(7)

		• Leverages a proven measure of usability • Standardizes your evaluations • Provides a manageable numeric score • Helps you make quick assessments.	
		Module II	
13	a)	Methods are: Stakeholder Mapping: • A way of diagramming the network of people who have a stake in a given system. • The balance of any ecosystem is defined by the interrelationships of its parts. • Each individual unit, whether key or peripheral, plays a role in how the system works. • As a visualization of people's relationships, interactions, and needs, a Stakeholder Map helps you understand the extent and impact of your design decisions. Persona Profile • An informed summary of the mindset, needs, and goals typically held by key stakeholders. • A good set of personas will guide you and your team to think about the recipients of your ideas at every stage of design development. • The best Persona Profiles are comprised of illustrations, descriptive text, and a photo portrait. • They can help your team remember and discuss the people for whom you are designing. • As a reference for generating ideas, prioritizing features, or discussing trade-off decisions, personas are a valuable tool for decision making. Experience Diagramming • A way of mapping a person's journey through a set of circumstances or task. • Experience Diagramming is an effective way to visualize your event-based research.	(8) 2 X 4

		• It is a great way to document the extent to which you understand the current state of a particular situation. • A useful diagram reveals more than just an overview of the action. • It can also bring to light important qualities of the experience, showing the complexity or difficulty people faced and how they struggled, adapted, or overcame. Concept Mapping • A way of depicting the relationships between various concepts in a given topic area. • Concept mapping as a technique for organizing concepts in a way that illustrates a thorough understanding of a topic, problem, or situation. • It uses diagrams to illustrate relationships between different concepts, typically by connecting them with labeled arrows. • Concept maps are widely used for learning, brainstorming, problem-solving, and presenting ideas.	
	b)	• A quad chart for plotting items by relative importance and difficulty. • A simple 2x2 matrix can be a powerful instrument for establishing priorities. • Specifically, placing Importance (low to high) on the x-axis and Difficulty (low to high) on the y-axis equips you to work out tensions between these opposing forces. • When you plot items according to both priorities, you and your team will likely arrive at a workable resolution. • The items that land in the lower left quadrant are characterized as targeted because they are the easiest to realize. • The upper left quad rant contains luxurious items—costly endeavors with little return. • The items in the upper right quadrant are considered to be strategic because they require large investments to get big results. • The items in the lower right quadrant are high-value because they yield great impact at a low price.	(6)

		Quick guide: • Identify a project that requires prioritization. • Make a poster showing a large quad chart. • Label horizontal axis Importance (or Impact). • Label vertical axis Difficulty (or Cost to Execute). • Form a team, and gather data for discussion. • Plot items horizontally by relative importance. • Plot items vertically by relative difficulty. • Consider the quadrants where items get placed. • Look for related groupings, and set priorities. Benefits: • Helps you prioritize items quickly • Facilitates deliberation • Resolves differing opinions • Helps your team develop a plan of action	
		OR	
14	a)	Methods are: Affinity Clustering • A graphic technique for sorting items according to similarity. • Sometimes in the midst of a project, an over whelming amount of information or ambiguity threatens to bog down the pace of progress. • Affinity Clustering will help you avoid this road block. • Whether analyzing research data or considering creative ideas, you can use this method to organize items into logical groups. • Helps you identify issues and insights. • Reveals thematic patterns. • Facilitates productive discussion. • Builds a shared understanding. Bull's-eye Diagramming • A way of ranking items in order of importance using a target diagram.	(8) 2 X 4

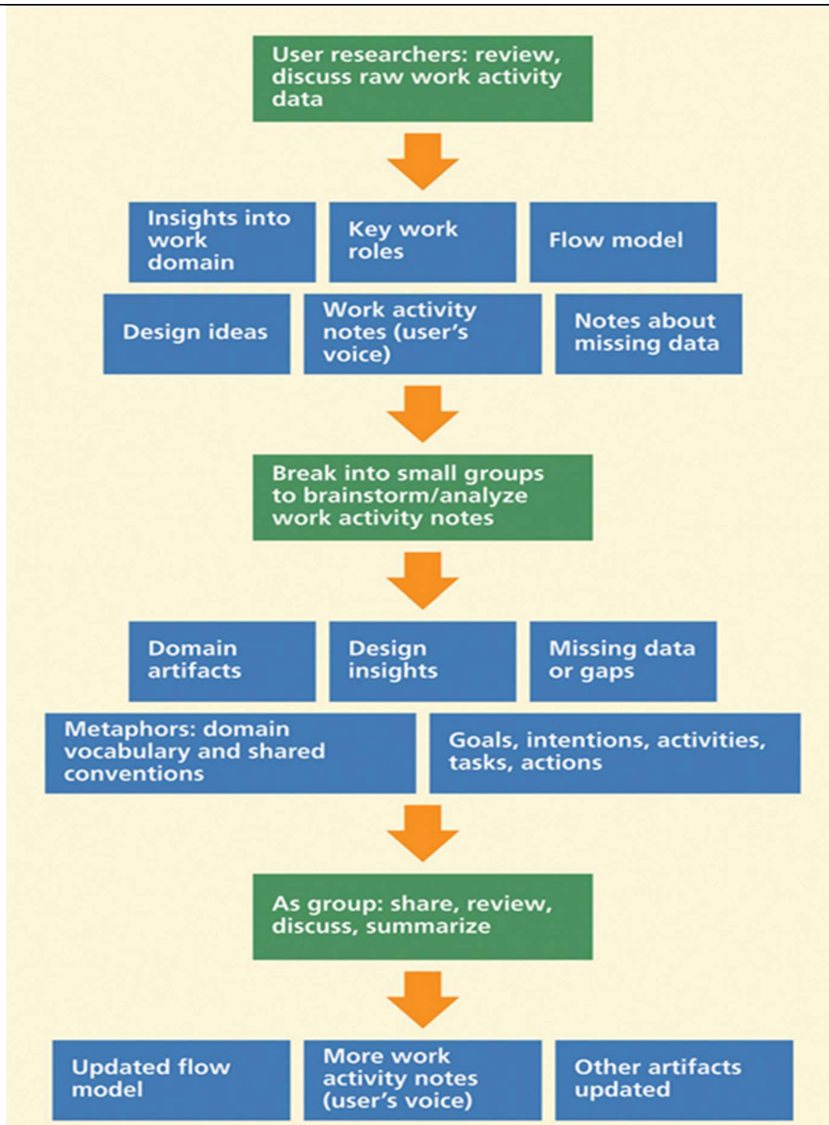
- Bull's-eye Diagram puts a limit on how much you can identify as critical, thus forcing your team to deliberate about essentials before producing anything.
- Since each successive circle is larger than the bull's-eye, you must carefully consider what is critical, what is important, and what is merely peripheral.
- Often this means having to make trade-off decisions.
- The result, however, is a clear delineation of your team's consensus about each item's relative importance.
- It is a comparatively simple method of making difficult decisions.
- Helps you determine what is most essential.
- Facilitates productive discussion Builds consensus.
- Helps your team develop a plan of action.

Importance/Difficulty Matrix

- A quad chart for plotting items by relative importance and difficulty.
- A simple 2x2 matrix can be a powerful instrument for establishing priorities.
- Specifically, placing Importance (low to high) on the x-axis and Difficulty (low to high) on the y-axis equips you to work out tensions between these opposing forces.
- When you plot items according to both priorities, you and your team will likely arrive at a workable resolution.
- The items that land in the lower left quadrant are characterized as targeted because they are the easiest to realize.
- The upper left quad rant contains luxurious items—costly endeavors with little return.
- The items in the upper right quadrant are considered to be strategic because they require large investments to get big results.
- The items in the lower right quadrant are high-value because they yield great impact at a low price.
- Helps you prioritize items quickly
- Facilitates deliberation

		• Resolves differing opinions • Helps your team develop a plan of action Visualize the Vote • A quick poll of collaborators to reveal preferences and opinions. • Visualize the Vote is a good way to get everyone's input, giving each person the opportunity to indicate preferences and opinions before final decisions are made. • Simple method to employ, and offers a good degree of flexibility. • When using it to pick the best solution among many, you can give everyone a token to cast a single vote. • Voting provides a quick catalyst for discussion, moving a project toward realization. • Helps you rate and rank preferences • Reveals thematic patterns • Diminishes overbearing opinions • Democratizes decision making	

	b)	• Interpretation of raw work activity data is accomplished through:
		    • building a flow model and
		    • synthesizing work activity notes
		• This flowchart illustrates a structured process for analyzing work activity data in the context of user research and design.
			(6)
			
			Fig: 2 mark
			Explain – 4 mark



- Here's a step-by-step breakdown of the process:
- **User Researchers: Review and Discuss Raw Work Activity Data (Green Box at the Top)**
    - Researchers gather and analyze raw data about how users perform their work activities.
- **Insights and Notes Generation (First Orange Arrow)**
    - From the raw data, researchers identify key components:
        - **Insights into the work domain**: General understanding of the field or area in question.

		• **Key work roles**: The roles or personas involved in the work. • **Flow model**: A high-level representation of how tasks and processes flow. • **Design ideas**: Early brainstorming ideas for improving or supporting work activities. • **Work activity notes (user's voice)**: Capturing users' input, feedback, and behaviors. • **Notes about missing data**: Identifying gaps in the information. • **Small Group Brainstorming/Analysis (Next Orange Arrow)** • Teams break into smaller groups to analyze the work activity notes collaboratively. • This involves brainstorming and identifying: • **Domain artifacts**: Documents, tools, or other items used in the work. • **Design insights**: Ideas or patterns relevant to potential solutions. • **Missing data or gaps**: Areas requiring further investigation or clarification. • **Group Outputs (Next Set of Blue Boxes)** • The group analysis leads to: • **Metaphors**: Establishing shared vocabulary and conventions to describe the domain. • **Goals, intentions, activities, tasks, actions**: A deeper understanding of the work structure and user motivations. • **Group Review, Discussion, and Summarization (Next Green Box)** • Teams reconvene as a whole to share their findings, discuss, and summarize insights. • **Updated Artifacts (Final Outputs, Blue Boxes at the Bottom)** • The final outcomes of the process include:	

		• **Updated flow model**: A refined representation of task flows. • **More detailed work activity notes (user's voice)**: Improved notes reflecting deeper understanding. • **Other artifacts updated**: Revisions to tools, documents, or other relevant materials. • This process emphasizes iterative collaboration, with user feedback and work context driving design insights and solutions.	
		Module III	
15	a)	**1.Maintain Connections to Your Data** • **Why**: Ensuring traceability between design decisions and raw contextual data enhances credibility and user-centeredness. • **How**: • Label all elements in the models with unique identifiers linked back to their original data sources. • Use line numbers, time codes, or WAAD (Work Activity Affinity Diagram) node IDs for reference. • Treat any item without a data reference as an unsupported assumption needing further scrutiny. **2. Extract Inputs Simultaneously with Requirements** • **Why**: This reduces redundancy and ensures consistency between requirements and models. • **How**: • While walking through the WAAD or other contextual data to extract requirements, simultaneously identify inputs for design-informing models. • Look for references to user types, tasks, social concerns, and other design-relevant elements in the data. • Take notes on potential model components as they emerge naturally. **3. Use Sorted Bins of Work Activity Notes** • **Why**:	(7) List 4: 1 mark Each one explain - 1.5 X 4

		• Pre-sorted data simplifies the synthesis of models and aligns with anticipated modeling needs.
		• **How:**
		  • Organize work activity notes into categories or "bins" during contextual analysis (e.g., user-related notes, task-related notes).
		  • Use these bins as inputs for creating corresponding models (e.g., personas, flow models, task descriptions).
		4. Represent Barriers to Work Practice
		• **Why:**
		  • Identifying barriers highlights opportunities for design improvements.
		• **How:**
		  • Include problems and interruptions observed in workflows (e.g., communication breakdowns, coordination issues).
		  • Use symbols like red lightning bolts (ϟ) to visually mark barriers in models.
		  • Address these barriers in the envisioned versions of the models.	

	b)	• Use a consistent structure with major categories, rationale, and traceability tags (e.g., WAAD node IDs). • Use headings, subheadings, and IDs to keep requirements organized and traceable. • Example: "Security: Shall protect ticket-buyer privacy [C19]". • A requirements document is essentially a set of requirement statements organized on headings at two or more levels. **Name of major feature or category** *Name of second-level feature or category* Requirement statement [WAAD source node ID] Rationale (if useful): Rationale statement Note (optional): Commentary about this requirement ***Generic structure of a requirement statement.*** • We use A, B, C, ... for the highest-level nodes under the root node.	(7) Structure - 2 mark Explain –3 mark Example: -2 mark

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- Under node A, we use AA, AB, AC, ...
- For the work activity notes themselves we use the group ID plus a number, such as AB1and AB2; this is the identifier that goes in the “WAAD source node ID” part of a requirement.

Security
Privacy of ticket–buyer transactions
Shall protect security and privacy of ticket-buyer transactions [C19]
Note: In design, consider timeout feature to clear screen between customers.

Example requirement statement.

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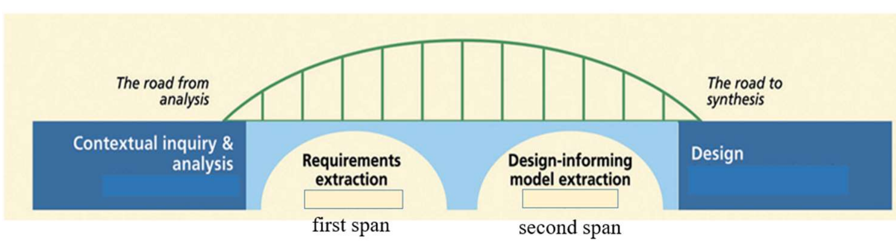
<tr><td></td><td></td><td>**OR**</td><td></td></tr>
<tr><td>16</td><td>a)</td><td>

Bridging the gap between Analysis and Design:

- Information coming from contextual studies describes the work domain but does not directly meet the information needs in design.
- The output of contextual inquiry and analysis does not speak directly to what is needed as inputs to design. There is a gap.
    - Information coming from contextual studies describes the work domain but does not directly meet the information needs in design.
    - There is a cognitive shift between analysis-oriented thinking on one side of the gap and design-oriented thinking on the other.
    - The gap is the demarcation between the old and the new—between studying existing work practice and existing systems and envisioning a new work space and new system design space.



The image uses a **bridge analogy** to represent the transition from understanding a problem to designing a solution.

<u>**1. The Road from Analysis:**</u>

- This is the starting point where the process focuses on **understanding the context and analyzing the problem**.

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First span – 3.5
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- Observing and interacting with users or stakeholders to gather insights about their behaviors, needs, and pain points.
- Analyzing this data to create a thorough understanding of the current situation.

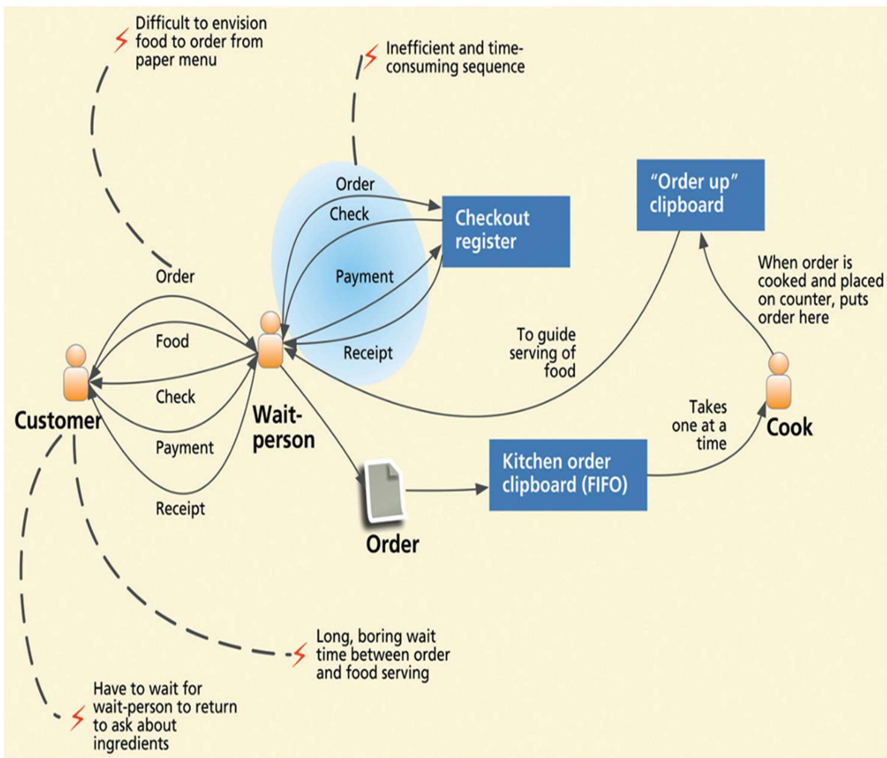
2. The Bridge:

- It is divided into two key components, which are **essential for translating insights into actionable outputs**:
    - **Requirements Extraction:** Identifies and organizes the **functional and non-functional requirements** that need to be addressed.
    - The term refers to **what is needed to design a system that will fulfill user and customer goals.**
    - In the UX domain, **interaction design requirements** describe what is required to support user or customer work activity needs.
    - We are also concerned with **functional requirements** to ensure the usefulness component of the user experience.

    - **Design-Informing Model Extraction: C**reating **abstract models or frameworks** based on the analysis to inform the design process.
    - The second span of the bridge in the transition from contextual analysis to design focuses on constructing design-informing models.
    - This stage involves transforming the insights derived from contextual inquiry and requirements extraction into actionable artifacts that guide and inspire the design process.
    - Design-informing models are constructs (e.g., personas, task descriptions) that transform raw data into elements that inspire and guide design decisions.

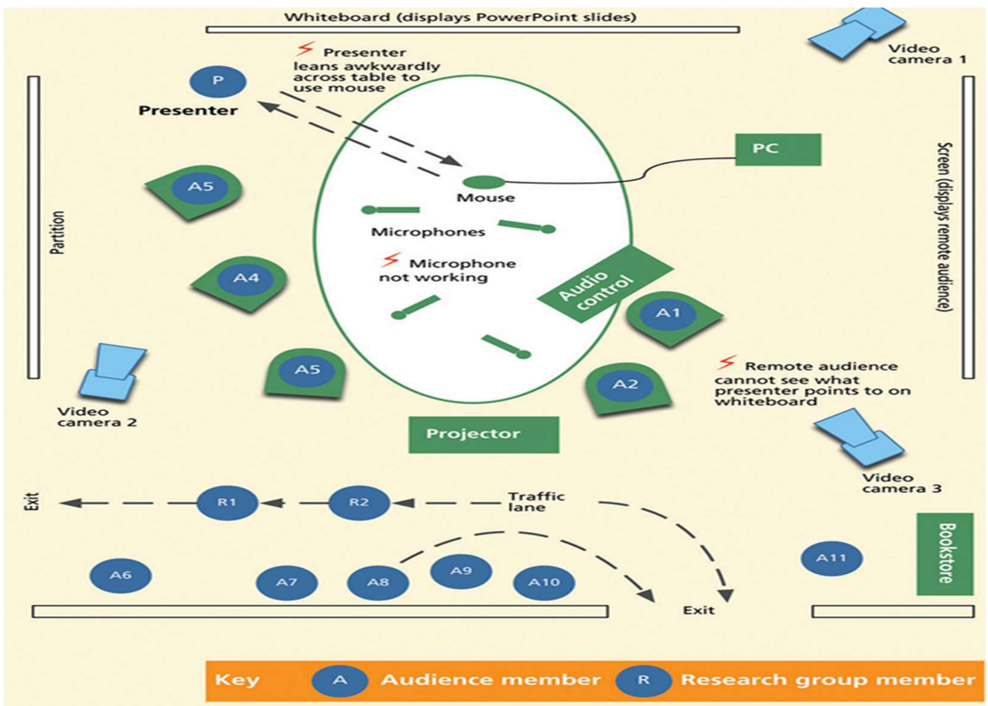
		• Design informing models serve as a bridge between user research and the actual design phase, ensuring that the final product effectively meets the needs and expectations of its users. **3. The Road to Synthesis:** • This phase represents **design and creation** based on the insights and models developed in the previous steps. • The focus here is on **solution-building**, and it is labeled as "Design." • Activities in this phase include: • Concept generation. • Prototyping and iteration. • Testing solutions to refine and validate them against user needs. • These two components (requirements and models) form the **pillars of the bridge**, enabling a smooth transition from understanding to ideating solutions.	
	b)	• This model describes the physical and contextual factors influencing how tasks are performed in a workspace. • It focuses on the environment where work occurs, capturing elements like physical constraints, tools, and workspace layout. • It includes: • Artifact Model • Physical model ❖ **Artifact Model** • This model represents the tangible elements—both physical and digital—that are used, shared, or modified during work activities. • These artifacts help reveal work practices and how information is handled during task completion. • **Key Components:**	(7) Model explain -1 mark Types: - 2 X 3 mark

- Artifacts: Tangible objects such as forms, documents, tools, or digital files (e.g., a restaurant menu, an order form, or a digital spreadsheet).
- Attributes: Properties of the artifact (e.g., fields in a form or labels on a chart).
- Usage Patterns: How the artifact is used, shared, or modified in the workflow.
- Eg: Part of a restaurant flow model with focus on work artifacts derived from the artifact model.



❖ **Physical Model**

- This model describes the physical layout and environment where tasks are performed.
- It shows how people, objects, and equipment are arranged within a workspace and how their physical positioning affects task efficiency and collaboration.
- **Key Components:**

		• Physical Space: The layout of rooms, desks, and physical structures. • Work Equipment: Devices like computers, printers, and telephones. • Artifact Placement: Locations of tools and documents related to tasks. • Movement Paths: How people move within the space, showing potential barriers or inefficiencies. • Eg: physical model for slideshow presentation 	
		Module IV	
17	a)	• The methods in this grouping help you compel others to take the necessary actions for great ideas to flourish. Concept Poster • A presentation format illustrating the main points of a new idea. • An effective Concept Poster is a powerful way to promote an idea and rally support for its development.	(7) List – 1 mark Each explain – 4 X 1.5

		• Start by thinking big—large scale, big ideas, short phrases—and then fill in the details to help support the main points you're highlighting. • Your poster should show what the idea is, why it matters, and how it works, to influence people to embrace your concept by communicating what makes it an appropriate solution. • Since this is a poster, you'll want to keep it highly visual, using images and words that clearly and concisely articulate the important aspects of your concept. • Remember that while words can aid understanding, illustrations, even if rudimentarily drawn, can go a long way in conveying an idea. • Promotes a vision of the future • Helps you build a business case • Gains support from decision makers • Provides a road map for moving forward Video Scenario • A short movie showing the attributes of a new concept in use. • IF A PICTURE is worth a thousand words, then a video is worth a thousand pictures. • You can spend great time and effort telling people about a new concept, but words only go so far—sometimes you have to show people the world you envision. • With video, you can present a compelling explanation of what your concept would be like in a true-to-life scenario. • This allows people to better imagine how one might interact with or experience your design. • As a vehicle for storytelling, a Video Scenario has the potential to capture your team's attention and gather their support for an idea before it is made real. • These days, creating, editing, and sharing video is common place and relatively easy.	

- While you may want someone with experience to polish the final product, you can do much of the groundwork yourself.
- Use your storyboarding skills to frame the narrative, and then use video to bring it to life!
- Helps people imagine the future
- Shows what a concept looks like in action
- Gains support from decision makers
- Inspires your team.

Cover Story Mock-up

- A mock news article describing the successful future of a new idea.
- It presents the idea as if it has already been successfully implemented and is being featured as a cover story in a leading publication.
- An idealized story is a unique way to explain the need or benefit of something by imagining the impact of its creation.
- If you are able to present your concept as a success, by showing its potential merits and a positive outcome, you are more likely to get buy-in from those who are initially uncertain about it.
- Your goal for a Cover Story Mock-up is to answer the question: What will this ___ have done?
- This requires you to consider the potential cultural impact of your idea and to frame its success around how people would ideally respond to it.
- When you begin with a positive ending, it’s easier to regroup people around the development of your idea.
- Shows a successful future state
- Promotes a shared vision
- Gains support from decision makers
- Inspires your team

Quick Reference Guide

- A short document summarizing the key principles and elements of a proposed solution.

		• A Quick Reference Guide ensures that your original design intent is maintained as the idea moves from conception through production and beyond. Since clarity is the goal, a good guide is essential. • Everything relevant to the design—style, form, composition, and conventions—is contained in one document that provides an orientation for those unfamiliar with the concept. Therefore, it must be easy to digest. • Begin your Quick Reference Guide as early in the process as you can. • This will offer you a chance to clearly articulate your team's design rationale as soon as you begin making decisions. • Then the document can evolve along with the project, eventually containing all of the information that defines the design. • When completed the document should answer these questions: What is it? What are the elements? How does it work? • Summarizes the rationale for your design • Promotes key principles • Provides specifications • Supports proper implementation	
	b)	• We describe three design perspectives as filters through which we view design and design representations to guide thinking, scoping, discussing, and doing design. ○ Ecological Perspective ○ Interaction Perspective ○ Emotional Perspective **Ecological Perspective** • The ecological design perspective is about how the system or product works within its external environment. • It is about how the system or product is used in its context and how the system or product interacts or communicates with its environment in the process. • This is a work role and workflow view, which includes social interaction and long-term phenomenological aspects of usage as part of one's lifestyle.	(7) List – 1 Each explain – 3 X 2

		• System infrastructure plays an important role in the ecological perspective because the infrastructure of a system, the other systems and devices with which it interacts in the world, is a major part of its ecology. Infrastructure leads you to think of user activities, not just isolated usage. • This perspective shifts the focus from simply producing a physical object to designing a product that provides a valuable service within a larger system, considering its environmental and social impact throughout its lifecycle. - "**A product is actually a service.**" **Emotional Perspective** • The emotional design perspective is about emotional impact and value-sensitive aspects of design. • It is about social and cultural implications, as well as the aesthetics and joy of use. • System infrastructure can also play a role in the emotional perspective because the infrastructure of a system provides scaffolding for the phenomenological aspects of usage, which are about broader usage contexts over longer periods of time. • A product is not just a product; it is an experience. • People do not usually use a product in isolation from other activities. • People use products as part of an activity, which can include many different kinds of usage of many different things. • And that starts with the out-of-the-box experience, which is not enhanced by difficult hard plastic encasing, large user manuals, complex installation procedures, and having to consent to a legal agreement that you cannot possibly read. **Interaction Perspective** • The interaction design perspective is about how users operate the system or product. • It is a task and intention view, where user and system come together. • It is where users look at displays and manipulate controls, doing sensory, cognitive, and physical actions.	
		OR	

18	a)	• Sketching is a rapid, freehand method for generating and exploring design ideas. • Sketching for Ideation: Sketching enhances creativity by providing a visual language for brainstorming, helping designers explore possibilities rather than refine finalized ideas. • Sketching vs. Prototyping: Unlike prototypes, which define and test decisions, sketches are ambiguous, tentative, and used for idea generation. • Embodied Cognition: Sketching is not just a representation of thought—it is an active part of thinking and problem-solving, engaging both mind and body. • Everyone can sketch; you do not have to be artistic. • Most ideas are conveyed more effectively with a sketch than with words. • Sketches are quick and inexpensive to create; they do not inhibit early exploration. • Sketches are disposable; there is no real investment in the sketch itself. • Sketches are timely; they can be made just-in-time, done in-the-moment, provided when needed. • Sketches should be plentiful; entertain a large number of ideas and make multiple sketches of each idea. • Textual annotations play an essential support role, explaining what is going on in each part of the sketch and how. **How Sketching Helps in Design Thinking** • Encourages Quick Idea Exploration: Sketches enable rapid iteration and brainstorming. • Supports Collaboration: Teams can annotate, modify, and discuss sketches together. • Serves as Design Documentation: Sketches preserve the evolution of ideas throughout a project. • Example: A sketch of an app's navigation flow helps designers visualize user interactions before coding begins.	(4) (3)
	b)	• A design paradigm is a way of thinking about and approaching design.	(1)

		• It's a set of ideas, methods, and assumptions that guide how designers solve problems.
		• Different design paradigms focus on different aspects of human-computer interaction (HCI), such as efficiency, cognition, or meaning-making.
		• Paradigms shape how designers understand users, create interfaces, and evaluate systems.
		• The different design paradigms are:
		    • Engineering Paradigm
		    • Human Information Processing (HIP) Paradigm
		    • Design-Thinking Paradigm
		1. Engineering Paradigm
		• The HCI Engineering Paradigm, drawing inspiration from software engineering and human factors, emphasizes a structured approach to designing user-friendly systems.
		• Begins by defining the desired system functionalities.
		• Prioritizes usability engineering, involving continuous evaluation and improvement to enhance user performance.
		• Employs a structured design process with regular evaluations and refinements.
		• Leverages human factors research to understand user needs, behaviors, and limitations.
		• Focuses on fulfilling user requirements and maximizing productivity.
		• Prioritizes efficiency and error reduction, often measured by user performance metrics.
		• It was a purely utilitarian and requirements-driven approach.
		2. Human Information Processing (HIP) Paradigm
		• The Human Information Processing (HIP) paradigm in HCI views the human mind and the computer as interconnected information processors.
		• HIP emphasizes understanding how humans perceive, process, and act upon information (sensing, cognition, memory, decision-making, and physical actions).
			3 X 2

		• By modeling human cognitive processes, designers can create interfaces that align with human capabilities and limitations. • The HIP paradigm has its roots in psychology and human factors, from which it gets an element of cognitive theory. • It is about modeling human sensing, cognition, memory, information understanding, decision making, and physical performance in task execution. • The idea was that once these human parameters were codified, it is possible to design a product that "matches" them. **3. Design-Thinking Paradigm** • Also called the "phenomenological matrix." • The Design-Thinking Paradigm is about designing for the experience, not just the product itself. • It considers the social, cultural, and emotional aspects of interaction. • Moves beyond traditional usability methods, which focus on refining existing designs rather than creating the right design from the start. • Uses participatory design to involve users in early prototypes and design exploration. • Shifts from evaluating single-task performance to considering emergent use and overlapping design phases (analysis, design, implementation, evaluation). • Focuses on emotional, cultural, and social aspects of interaction, incorporating ideas from psychology, anthropology, and technology. • Considers "embodied interaction", where interaction involves the whole body and spirit, not just traditional input devices. • "Situated design" emphasizes the role of place in human-technology interaction. • Promotes joy, aesthetics, and a deep connection to products beyond mere functionality.	
		Module V	
19	a)	• Fidelity refers to **how closely a prototype resembles the final product** in terms of appearance, functionality, and interaction.	(1)

		• It reflects how "finished" a prototype looks, though it doesn't necessarily mean the underlying functionality is complete.
		• Types of Prototype Fidelity”
		  ○ Low-Fidelity Prototypes
		  ○ Medium-Fidelity Prototypes
		  ○ High-Fidelity Prototypes
		  ○ Low-Fidelity Prototypes
		Low-Fidelity Prototypes
		Characteristics:
		• Simple, rough representations of the design.
		• Usually made using paper, sketches, or basic tools.
		• Do not include detailed look, feel, or behavior.
		Advantages:
		• Easy to create, modify, and iterate upon.
		• Inexpensive and quick to produce.
		• Effective for brainstorming and early design exploration.
		• Encourages team collaboration and creativity.
		• Focuses feedback on **high-level design issues** like workflows and functionality, rather than aesthetics.
		Misconceptions:
		• Some teams may initially dismiss low-fidelity prototypes as "amateurish," but studies have shown users take them seriously and provide valuable feedback.
		Use Cases:
		• Early design stages to test concepts and workflows.
		• Quickly gathering user feedback.
		Medium-Fidelity Prototypes
		Characteristics:
		• More polished than low-fidelity prototypes but not fully realistic.
		• Often computer-based, such as wireframes (basic visual guide or blueprint that represents the structure and layout of a digital product) or simple digital mockups.
			4 X 1.5

		● Demonstrate **layout, navigation, and basic interactions**. **Advantages**: ● Balance between detail and speed of production. ● Suitable for testing interaction design and navigation flows. ● Useful for communicating with stakeholders and development teams. **Use Cases**: ● Intermediate design stages when you need more detailed feedback but don't want to commit to final design details. **High-Fidelity Prototypes** **Characteristics**: ● Detailed, realistic representations of the final product. ● Include specific appearance, behavior, and interaction details. ● May include functional back-end components (e.g., working databases or APIs). **Advantages**: ● Allow users to evaluate the design as if it were the real product. ● Useful for refining detailed design elements like color schemes, animations, and error handling. ● Serve as effective tools for stakeholder presentations, marketing demos, or securing investment. **Disadvantages**: ● Expensive and time-consuming to create. ● Difficult to modify, especially if design changes are required. **Use Cases**: ● Later stages of design to finalize appearance and interactions. ● Advanced evaluations and usability testing.	
	b)	● This part describes the moment when the project moves from **prototyping** to actual **software development**. The UX team's work of refining interaction design comes to a close, and the **software engineering (SE) team takes over**.	(7)

- Once the UX process ends, the project **shifts focus to final implementation**.
- **Some formative evaluation can still be done**, but now it is with the **real software system** instead of the prototype.
- **Further changes become more expensive and time-consuming**, making it crucial that the prototype was refined properly.
- The UX team may **either disband or move on to a new project** while the development team finalizes the product.

The transition from a prototype to a final product raises an important question: **Should the prototype code be reused or discarded?**

<u>You Cannot Just Keep the Prototype</u>

- While a prototype **may look functional**, it is **not built for real-world use**.
- Prototypes are created quickly, often **without considering performance, security, or maintainability**.
- The prototype **may have been built using tools or platforms unsuitable** for the final product.

<u>How Do You Reuse the Interaction Design of the Prototype?</u>

- The final product should **faithfully follow the prototype's interaction design** to maintain the user experience.
- The **layout, user flows, and interface elements should remain intact**, even if the underlying code is rewritten.
- Small changes in **font size, button placement, or menu structure** can **negatively impact usability**.

<u>Using a Prototype as a Design Specification</u>

- Many teams use **high-fidelity prototypes** as **interaction design specifications** for developers.
- A **project style guide** (covering colors, fonts, branding, and UI rules) helps ensure

design consistency.

- Developers must carefully translate the prototype into production software **without losing essential design details**.

		The Need for UX and SE Collaboration and Respect The transition from prototype to product requires **close collaboration between UX designers and software engineers**. **Key challenges in the transition:** 1. **UX teams "own" the interaction design, but not the implementation code.** 2. SE teams may **modify the design to fit technical constraints**, sometimes altering the user experience. 3. **UX designers must actively ensure that SE teams follow the intended design**, since SE teams may not be experts in user experience. 4. **Mutual respect and collaboration** are essential to ensure the final product stays true to the prototype's usability goals. ***Do Not Think the UX Team Is Now Done*** ● Just because the SE team takes over does **not** mean the UX team's job is finished. ● There are **no universal software development standards** that ensure **Interaction design is implemented correctly**. ● Developers may **unintentionally change interaction details**, affecting usability. ● UX designers should **stay involved** during implementation to **review and test the final product** for consistency with the prototype.	
		OR	
20	a)	**1. Scripted and "Click-Through" Prototypes** **Scripted Prototypes**: • Programmed using simple scripting languages to add basic behaviors. • Create "live-action storyboards" that respond in a limited way to user actions. • Typically low- or medium-fidelity. • Examples: Moving between predefined screens or animations triggered by specific actions. **Click-Through Prototypes**:	(9)

- Contain active links or buttons that allow users to navigate through screens.
- Often created using wireframes or mockups with clickable elements.
- Limited interactivity: clicking moves the user to the next screen or section but lacks complex functionality.
- Examples: Clicking a "Next" button to navigate through screens in a user flow.

2. Fully Programmed Prototypes

Characteristics:

- High-fidelity prototypes with a significant level of functionality.
- Simulate the complete behavior of the system, often including back-end processes (e.g., a working database).
- Expensive and time-consuming to create but provide a highly realistic experience.

Use Case:

- Used in large projects where stakeholders or users need to experience the product in detail before full implementation.
- Example: An air traffic control system prototype built as part of a large-scale project proposal.

3."Wizard of Oz" Prototypes

- A technique where the user interacts with what appears to be an automated system, but a human is controlling the responses behind the scenes.
- **Setup**:Two computers are connected: one for the user and one for a hidden evaluator who manually simulates system responses.

Advantages:

  - Simulates highly interactive systems, even those requiring complex or adaptive behavior (e.g., AI-driven applications), without full implementation.
  - Useful for exploring user behavior in uncertain or evolving scenarios.

Example:

		○ Early user interface designs for an email system where a human simulated system responses to unknown commands, allowing iterative improvement of the design. **4. Physical Mockups** **Physical mockups** are prototypes created for systems or products with a physical component, such as handheld devices, kiosks, or other hardware. **Approach**: ○ Can involve materials like cardboard, wood, or even 3D-printed components to mimic the form factor and interaction. ○ Enhances realism by allowing users to interact with the physical design. **Example**: ○ A cardboard prototype of a mobile device used to test ergonomics and button placement.	
	b)	Storyboards typically show **static frames**, but **user experience is about transitions** between those frames. ● **Transitions define how interactions feel**, such as: ○ Screen fades, animations, or button presses. ○ How quickly a response appears. ○ What happens **between** selecting an option and seeing the next screen. ● Example: A **storyboard for a mobile app** should show: ○ User **scrolling through options** (gesture). ○ A **loading animation** while the system processes the request. ○ A **smooth transition** to the next screen. **Why Are Transitions Important?** ● Transitions **shape the user's perception of the system**. ● Poorly designed transitions create **confusion and frustration**. ● Good transitions improve **usability and satisfaction**.	(5)
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